

Prevalence of Kidney Stones in the United States



Background The last nationally representative assessment of kidney stone prevalence in the United States occurred in 1994. After a 13-yr hiatus, the National Health and Nutrition Examination Survey (NHANES) reinitiated data collection regarding kidney stone history. **Objective** Describe the current prevalence of stone disease in the United States, and identify factors associated with a history of kidney stones. **Design, setting, and participants** A cross-sectional analysis of responses to the 2007–2010 NHANES (n = 12 110). **Outcome measurements and statistical analysis** Self-reported history of kidney stones. Percent prevalence was calculated and multivariable models were used to identify factors associated with a history of kidney stones. **Results and limitations** The prevalence of kidney stones was 8.8% (95% confidence interval [CI], 8.1–9.5). Among men, the prevalence of stones was 10.6% (95% CI, 9.4–11.9), compared with 7.1% (95% CI, 6.4–7.8) among women. Kidney stones were more common among obese than normal-weight individuals (11.2% [95% CI, 10.0–12.3] compared with 6.1% [95% CI, 4.8–7.4], respectively; $p < 0.001$). Black, non-Hispanic and Hispanic individuals were less likely to report a history of stone disease than were white, non-Hispanic individuals (black, non-Hispanic: odds ratio [OR]: 0.37 [95% CI, 0.28–0.49], $p < 0.001$; Hispanic: OR: 0.60 [95% CI, 0.49–0.73], $p < 0.001$). Obesity and diabetes were strongly associated with a history of kidney stones in multivariable models. The cross-sectional survey design limits causal inference regarding potential risk factors for kidney stones. **Conclusions** Kidney stones affect approximately 1 in 11 people in the United States. These data represent a marked increase in stone disease compared with the NHANES III cohort, particularly in black, non-Hispanic and Hispanic individuals. Diet and lifestyle factors likely play an important role in the changing epidemiology of kidney stones.